

EV BATTERY PERFORMANCE AND RANGE MONITORING SYSTEM

*A Project report submitted in partial fulfilment of the requirements for the Award of the
degree of*

BACHELOR OF TECHNOLOGY

In

CSE- (Artificial Intelligence & Machine Learning)

Submitted

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DECLARATION

We here by declare that the project dissertation entitled “**EV BATTERY PERFORMANCE AND RANGE MONITORING SYSTEM**” submitted for the B.Tech. degree is our original work and the dissertation has not formed the basis for the award of any degree, associateship, fellow- ship, or any other similar titles.

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ABSTRACT

The EV Battery Performance and Range Monitoring System is developed to analyze electric vehicle charging behavior and driving range using datasets collected from both Indian and global sources. With the rapid adoption of electric vehicles, understanding battery performance and range variability has become crucial for improving user confidence and optimizing vehicle efficiency. This system focuses on monitoring key battery parameters and analyzing real-world usage patterns to provide meaningful insights into EV performance under different operating conditions.

The project examines the relationship between battery capacity, charging patterns, vehicle efficiency, and driving range through a data-driven approach. By integrating historical and real-time data, the system identifies how factors such as charging frequency, depth of discharge, temperature variations, and driving habits influence battery health and vehicle range. This analysis helps in recognizing performance degradation trends and predicting future battery behavior, which is essential for effective battery management and long-term vehicle reliability.

To ensure accurate and reliable analysis, the collected datasets undergo comprehensive data cleaning and preprocessing steps. Noise removal, handling of missing values, normalization, and feature extraction techniques are applied to prepare the data for meaningful interpretation. Advanced analytical methods and visualization tools are then used to identify patterns, correlations, and anomalies in battery performance. These insights are presented through interactive dashboards that display key metrics such as charging time, energy consumption, battery efficiency, and estimated driving range in an intuitive and user-friendly format.

Overall, the developed system plays a significant role in reducing range anxiety by providing transparent and data-backed information about battery performance and vehicle capability. It supports EV users in planning trips more efficiently, assists researchers in studying battery behavior, and helps policymakers in understanding real-world EV usage trends for infrastructure planning. By combining data analytics with visualization, the project contributes to smarter energy utilization, improved battery lifecycle management, and the broader adoption of sustainable electric mobility.

In addition to its current capabilities, the system can serve as a foundation for integrating intelligent decision-support mechanisms in the future. By incorporating predictive analytics and real-time data streams from connected vehicles and charging networks, the platform can evolve into a proactive monitoring system that not only analyzes past performance but also provides recommendations for optimal charging strategies and energy usage. For instance, users could receive suggestions on the best time and location to charge their vehicles based on battery status, traffic conditions, and charging station availability. Furthermore, the system can support advancements in smart grid integration by aligning EV charging behavior with energy demand patterns, thereby promoting efficient energy distribution. Such enhancements would strengthen the role of the system in advancing sustainable transportation, improving battery utilization, and enabling data-driven decision-making across the electric mobility ecosystem.

CHAPTER 1

INTRODUCTION

CHAPTER 1: INTRODUCTION

1.1 Background

In recent years, the global transition toward sustainable transportation has accelerated the adoption of electric vehicles (EVs) as an alternative to conventional fuel-powered vehicles. Governments, automotive manufacturers, and environmental organizations are promoting EV usage to reduce greenhouse gas emissions, lower dependence on fossil fuels, and support clean energy initiatives. As EV deployment increases across urban and rural regions, the performance, reliability, and efficiency of EV batteries have become critical factors influencing user acceptance and large-scale adoption.

The battery is the most vital and expensive component of an electric vehicle, directly affecting driving range, charging time, vehicle performance, and overall lifecycle cost. However, EV users often face uncertainties related to battery degradation, range variability under different driving conditions, and charging efficiency. These uncertainties contribute to **range anxiety**, which remains one of the primary barriers to widespread EV adoption. Monitoring and analyzing battery behavior in real-world conditions is therefore essential for improving reliability, enhancing user confidence, and optimizing vehicle performance.

Traditional vehicle monitoring systems provide only basic information such as battery percentage and estimated range, without offering deeper insights into battery health, performance trends, or energy consumption patterns. With the increasing availability of vehicle telemetry data and advancements in data analytics, there is a growing need for intelligent systems that can process large volumes of EV data to provide meaningful, real-time insights into battery performance and driving efficiency.

This scenario highlights the importance of developing a comprehensive EV Battery Performance and Range Monitoring System that can analyze historical and real-time data, detect performance anomalies, and assist users, manufacturers, and policymakers in making informed decisions regarding battery usage, charging infrastructure, and energy optimization.

1.2 Problem Statement

Despite the rapid growth of electric mobility, several technical and operational challenges hinder the effective monitoring and optimization of EV battery performance:

- **Battery Degradation and Health Monitoring:** EV batteries degrade over time due to repeated charging cycles, temperature fluctuations, and driving patterns. Without proper monitoring, it becomes difficult to predict battery lifespan and maintenance requirements.
- **Range Uncertainty:** Driving range varies significantly depending on factors such as road conditions, driving style, climate, battery age, and load. This unpredictability creates anxiety among EV users and complicates trip planning.
- **Lack of Real-Time Performance Insights:** Many existing EV systems provide limited real-time analytics, preventing users from understanding how their driving and charging behaviors affect battery efficiency and energy consumption.

- **Data Complexity and Volume:** Modern EVs generate large volumes of sensor and telemetry data, including voltage, current, temperature, state of charge, and energy usage. Processing and analyzing this data efficiently requires advanced data handling and visualization techniques.
- **Need for Data-Driven Decision Making:** Stakeholders such as EV users, manufacturers, and policymakers require accurate and interpretable insights to improve battery management strategies, optimize charging infrastructure, and support sustainable transportation planning.

These challenges necessitate the development of a smart, data-driven monitoring system that can track battery performance, analyze energy consumption patterns, and provide reliable range predictions. The proposed EV Battery Performance and Range Monitoring System addresses these issues by integrating data preprocessing, analytical modeling, and interactive dashboards to deliver actionable insights and improve overall electric vehicle efficiency and user experience.

1.3 Project Objectives

The primary objective of this project is to design and develop an intelligent EV Battery Performance and Range Monitoring System that analyzes battery behavior, energy consumption, and driving patterns to improve vehicle efficiency and provide reliable range insights. The system aims to support EV users, researchers, and stakeholders by delivering accurate, real-time information about battery health and performance. The specific objectives of the project are:

- **Real-Time Battery Monitoring:** Develop a system capable of continuously tracking key battery parameters such as state of charge, voltage, current, temperature, and energy consumption to provide real-time performance insights.
- **Accurate Range Estimation:** Design analytical models that estimate the remaining driving range of an electric vehicle based on battery condition, driving behavior, and environmental factors, helping users plan trips more effectively.
- **Efficiency Analysis and Optimization:** Evaluate energy usage patterns under different driving conditions to identify inefficiencies and suggest strategies for improving overall vehicle energy efficiency.
- **Battery Health Assessment:** Monitor battery degradation trends over time and provide insights into battery lifespan, charging cycles, and performance deterioration to support predictive maintenance and cost planning.
- **Data-Driven Visualization:** Develop interactive dashboards and visual analytics tools to present battery performance, charging patterns, and range predictions in an intuitive and easily interpretable format.
- **Scalable Data Processing:** Design the system to handle large volumes of vehicle telemetry data efficiently, ensuring reliable performance as the number of monitored vehicles or datasets increases.
- **Support for Sustainable Mobility:** Provide actionable insights that contribute to improved battery utilization, reduced energy wastage, and enhanced user confidence, thereby promoting the adoption of electric vehicles and sustainable transportation.

1.4 Project Scope

The scope of this project focuses on the design and development of a data-driven EV Battery Performance and Range Monitoring System that analyzes electric vehicle battery behavior, energy consumption, and driving efficiency. The project emphasizes data collection, processing, analysis, and visualization to provide meaningful insights into battery health and vehicle range performance. The project includes:

- **System Architecture Design:** Designing a structured and modular system architecture that supports data acquisition, storage, processing, and visualization for monitoring EV battery parameters and performance metrics.
- **Data Collection and Processing:** Gathering EV-related datasets containing parameters such as battery capacity, state of charge, charging duration, energy consumption, and driving distance, followed by preprocessing steps including data cleaning, normalization, and feature extraction.
- **Performance and Range Analysis:** Developing analytical models to evaluate battery efficiency, estimate driving range, and identify performance trends under different driving and environmental conditions.
- **Visualization and Dashboard Development:** Creating interactive dashboards and graphical representations to display key metrics such as battery health, energy usage, charging patterns, and estimated range in an intuitive and user-friendly manner.
- **Battery Health and Efficiency Monitoring:** Implementing techniques to track battery degradation, charging cycle patterns, and efficiency variations to support predictive maintenance and improved battery management.
- **Testing and Validation:** Conducting testing using real-world and simulated datasets to verify the accuracy of range predictions, performance analysis, and system stability, along with documenting system functionality and results. However, to maintain a clear and manageable project boundary, certain elements are excluded from the scope:
- **Hardware-Level Battery Management System (BMS) Development:** The project focuses on data analysis and monitoring rather than designing or modifying the physical battery management hardware within the vehicle.
- **Real-Time Vehicle Control Integration:** Direct integration with vehicle control systems such as motor controllers or onboard diagnostics for real-time actuation is not included in this implementation.
- **Development of New Battery Technologies:** The project does not involve the design, chemical modeling, or manufacturing of battery cells, as it is centered on monitoring and analytical evaluation of existing battery systems.
- **Mobile Application Deployment:** While dashboards are developed for analysis and visualization, full-scale deployment as a commercial mobile or embedded vehicle application is beyond the current scope.

This well-defined scope ensures that the project remains focused on analyzing EV battery performance and driving range, while providing a scalable foundation that can be extended in future work to support real-time vehicle integration, advanced predictive analytics, and smart energy management systems.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2: LITERATURE SURVEY

2.1 Evolution of Electric Vehicle Monitoring Systems

The monitoring of electric vehicle (EV) performance has evolved significantly alongside advancements in battery technology, vehicle connectivity, and data analytics. Early EV systems provided only basic information such as battery percentage and estimated range through simple onboard displays. These first-generation systems lacked the ability to analyze historical performance or provide predictive insights, limiting their usefulness for long-term battery management.

With the development of onboard diagnostics and telematics in the early 2000s, second-generation monitoring systems introduced more detailed performance tracking. These systems began collecting parameters such as battery voltage, current, and temperature, enabling manufacturers to analyze battery behavior during operation. However, the data was often stored locally and processed offline, making real-time monitoring and remote analysis difficult.

The third generation of EV monitoring systems emerged with the integration of wireless connectivity, cloud storage, and IoT technologies. Vehicles were able to transmit real-time telemetry data to centralized platforms, allowing remote diagnostics, fleet monitoring, and improved maintenance planning. Despite these improvements, many systems still focused primarily on fault detection rather than detailed efficiency and range analysis.

The current generation of EV monitoring systems integrates advanced data analytics, machine learning, and interactive visualization tools to provide comprehensive insights into battery performance and driving efficiency. These systems are capable of predicting range, identifying degradation patterns, and optimizing energy consumption. The shift from basic monitoring to intelligent, data-driven analysis marks a significant step toward improving reliability, reducing range anxiety, and supporting large-scale EV adoption.

2.2 Data Analytics and Machine Learning in EV Performance Analysis

Data analytics and machine learning techniques have played a crucial role in improving the accuracy of battery performance monitoring and driving range prediction. Traditional analytical methods relied on simple mathematical models and fixed efficiency assumptions, which often failed to capture the dynamic behavior of batteries under real-world conditions.

Machine learning models have been increasingly applied to predict battery state of health (SoH), remaining useful life (RUL), and driving range based on historical and real-time data. Various algorithms have been explored in literature for EV battery analysis:

Technique	Advantages	Limitations
Linear Regression	Simple and interpretable; useful for baseline range estimation	Limited in modeling non-linear battery behavior
Decision Trees	Easy to understand and handle multiple input features	May overfit without proper pruning
Random Forest	High prediction accuracy and robustness	Increased computational complexity
Neural Networks	Effective in modeling complex battery degradation patterns	Requires large datasets and difficult to interpret
Gradient Boosting	High performance in range prediction tasks	Requires careful tuning and higher computation

Recent studies show that ensemble learning methods provide improved prediction accuracy compared to single models by combining the strengths of multiple algorithms. These approaches are particularly useful in handling complex and non-linear relationships between battery parameters, environmental conditions, and vehicle performance. Machine learning-based systems also enable adaptive learning, allowing models to improve over time as more driving and charging data becomes available.

2.3 Real-Time Data Processing and Visualization in EV Systems

Real-time data processing is essential for monitoring battery performance and providing accurate range predictions during vehicle operation. Modern EV monitoring systems rely on continuous data streams from sensors measuring voltage, current, temperature, speed, and energy consumption. Processing this data in real time allows the system to detect abnormal behavior, update range estimates dynamically, and alert users to potential battery issues.

Several architectural approaches have been proposed for real-time EV data processing, including stream processing pipelines, edge computing, and event-driven system design. Stream processing frameworks enable continuous ingestion and analysis of telemetry data, ensuring that performance insights are updated without delay. Edge computing techniques further enhance responsiveness by performing initial data processing within the vehicle before transmitting summarized data to cloud servers.

Visualization plays a critical role in transforming complex battery and vehicle data into meaningful insights. Interactive dashboards, graphs, and real-time indicators help users understand energy usage trends, charging efficiency, and remaining driving range in an intuitive manner.

CHAPTER 3

SYSTEM ANALYSIS AND FEASIBILITY STUDY

CHAPTER 3: SYSTEM ANALYSIS AND FEASIBILITY STUDY

3.1 Existing System

In the electric vehicle domain, information related to battery performance, driving range, charging infrastructure, and vehicle efficiency is often scattered across multiple platforms such as manufacturer websites, government portals, and third-party databases. Users typically rely on manual comparison and fragmented sources to evaluate electric vehicles, which makes analysis time-consuming and less reliable.

1. Manual Comparison of EV Specifications

Consumers and analysts manually compare EV specifications such as battery capacity, range, charging time, and price. This process is inefficient and prone to errors due to inconsistent data formats and incomplete information.

2. Static Data Representation

Most EV platforms provide static tables or basic charts without interactivity. Users cannot dynamically filter or explore relationships between parameters like battery capacity, range, and efficiency.

3. Limited Insights into Charging Infrastructure

Existing systems may list charging station locations but often lack region-wise distribution analysis, station type categorization, or interactive maps, limiting strategic planning and user understanding.

4. Lack of Integrated Analysis Platform

There is no unified system combining EV specifications, charging station data, and performance analytics for comparative and analytical studies.

Limitations of Existing Systems

- Dependence on manual data comparison
- Lack of centralized EV data repository
- Limited interactive visualization capabilities
- Difficulty in identifying relationships between battery capacity and range
- Absence of analytical dashboards for performance monitoring
- No integrated system combining vehicles and charging infrastructure data

3.2 Proposed System

The proposed system introduces a data-driven platform that integrates electric vehicle datasets, charging station information, and interactive visual analytics using Tableau. The system is designed to provide clear insights into EV battery **performance, range efficiency, vehicle pricing, and charging infrastructure distribution.**

Key Components and Features

1. **Data Visualization:** Interactive Tableau dashboards display EV specs, brand comparisons, charging stations, and efficiency analysis.
2. **Centralized Dataset:** Multiple EV datasets are combined into a structured database for unified analysis.
3. **Geographical Analysis:** Map-based visuals show charging station distribution across India.
4. **Comparative Vehicle Analysis:** Users can compare EVs by battery, speed, range, body style, and powertrain.
5. **Flask Web Interface:** A browser-based Flask application hosts all dashboards for centralized access.
6. **Integrated Dashboard & Story:** Tableau dashboards and stories are embedded for interactive data exploration.

Key Advantages of Proposed System

- Centralized platform for EV data analysis
- Improved understanding through interactive dashboards
- Reduced manual effort in comparing EV specifications
- Visual insights into battery performance and range efficiency
- Better planning insights using charging station distribution maps
- Scalable architecture for incorporating future datasets

3.3 Methodologies

The development of this project follows a structured data analytics and visualization methodology to ensure accurate analysis and reliable insights.

1. **Data Collection:** EV datasets were gathered from public sources, including battery capacity, range, charging time, price, and vehicle specifications, along with charging station locations in India.
2. **Data Preprocessing:** Datasets were cleaned by removing duplicates, correcting inconsistencies, and handling missing values to ensure accuracy.
3. **Exploratory Data Analysis (EDA):** Initial analysis was performed to understand battery distribution, range variations, pricing trends, and charging station availability.
4. **Data Visualization:** Tableau dashboards and maps were created to represent EV data in an easy-to-understand visual format.
5. **Feature-Based Analysis:** Key parameters such as battery capacity, efficiency, charging type, and price were analyzed to study their impact on range and performance.
6. **Dashboard Integration:** Tableau dashboards were embedded into a Flask web application for an interactive, user-friendly interface.

3.4 Software Development Cycle

The project was developed using a structured and iterative development cycle to ensure systematic progress and quality output.

1. Requirement Analysis

- Need: Centralized platform to track EV battery health and range efficiency.
- Users: Individual buyers, fleet operators, and urban planners.

2. System Design

- Components: PostgreSQL DB, Python (Flask) backend, Tableau dashboards, HTML/CSS frontend.

3. Data Preparation

- Sample raw data:
 - EV records with missing range or battery capacity.
 - Charging-station records with duplicate entries for the same location.

4. Visualization Development

- Example dashboard elements:
 - Bar chart: “Top 5 EV brands by average efficiency (kWh/km).”

5. Web Application Development

- Example Flask routes:
 - `/dashboard` → serves embedded Tableau view.
 - `/predict` → accepts JSON `{model, battery_age, climate}` and returns estimated range.

6. System Integration

- Example integration:
 - Tableau dashboard URL: `https://public.tableau.com/.../ev_trends`.
 - Embedded in Flask using an `<iframe>` or Tableau JavaScript API.

7. Testing

- Example test cases:
 - “Dashboard loads within 3 seconds on Chrome (desktop).”
 - **“Range prediction changes correctly when battery age increases.”**

8. Deployment

- Example deployment steps:
 - Run Flask app on localhost or a server (e.g., `gunicorn app:app`).

9. Maintenance

- Example maintenance activities:
 - Monthly update of EV model list and charging-station dataset.
 - Quarterly refresh of prediction models using new real-world driving logs.
 - Bug fixes and UI improvements based on user feedback.

3.5 System Analysis – Stakeholder Analysis

Stakeholders

- Electric Vehicle Consumers: Need clear insights into battery performance and range before purchasing an EV.
- Fleet Operators: Require comparative analysis to select efficient vehicles for commercial operations.
- Government and Urban Planners: Use charging station distribution data to plan infrastructure development.
- Automotive Researchers and Analysts: Study EV performance trends and technological improvements.

Functional Requirements

- Import and store EV and charging station datasets
- Visualize EV specifications using dashboards
- Display charging station locations using interactive maps
- Provide filtering and comparison of vehicles based on range, battery capacity, and price
- Embed dashboards into a web-based user interface

Non-Functional Requirements

- High system responsiveness when loading dashboards
- Accurate and consistent data representation
- User-friendly interface for non-technical users
- Scalability to support future EV datasets
- Reliable performance during multiple dashboard interactions

Data Flow Analysis

1. Data collection from EV and charging station sources
2. Data cleaning and preprocessing

3. Storage in structured database or files
4. Data visualization using Tableau
5. Dashboard embedding into Flask web interface
6. User interaction with dashboards and filters
7. Display of analytical insights and comparisons

3.6 Feasibility Study

Technical Feasibility

The proposed system is technically feasible as it utilizes well-established and widely supported technologies such as Python, Tableau, SQL, and Flask. Python provides powerful libraries like Pandas and NumPy for efficient data processing and analysis. Tableau offers advanced data visualization capabilities, enabling the creation of interactive dashboards. SQL ensures reliable data storage and management, while Flask enables seamless web integration. These tools are compatible with standard computing environments and require no specialized hardware, ensuring smooth development, deployment, and maintenance of the system. Additionally, the system can be scaled or enhanced in the future with minimal technical constraints.

Economic Feasibility

The project is economically viable as it primarily depends on open-source tools and publicly available datasets. Python and Flask are free to use, and Tableau provides academic licenses, reducing overall software costs. The system does not require high-end hardware, further minimizing expenses. Since the project focuses on software-based implementation, operational and maintenance costs are also low. This makes the solution cost-effective for academic purposes and scalable for real-world applications without significant financial investment.

The system is designed with a user-friendly interface that allows users to easily interact with dashboards and visualizations. End users can access the system through a web browser without needing advanced technical knowledge. The dashboards include intuitive filters, charts, and graphical representations, making it easy to interpret EV battery performance data. The system also supports real-time or near real-time data updates, improving usability and efficiency. Overall, the system can be smoothly integrated into practical environments with minimal training requirements.

Schedule Feasibility

- Phase 1 (1–2 months): Data collection, dataset cleaning, and database preparation
- Phase 2 (1–2 months): Dashboard development and exploratory data analysis using Tableau
- Phase 3 (2–4 weeks): Flask web integration and system testing
- Total Timeline: Approximately 4–5 months with iterative development and validation

CHAPTER 4

DATA ACQUISITION

CHAPTER 4: DATA ACQUISITION

4.1 Collection of the Dataset

The development of the EV Battery Performance and Range Monitoring System required reliable and diverse datasets that represent real-world electric vehicle specifications, battery characteristics, and charging infrastructure. For this purpose, multiple publicly available datasets were collected and combined to form a comprehensive data source capable of supporting range prediction and efficiency analysis.

The primary dataset contains technical specifications of electric vehicles, including battery capacity, driving range, efficiency ratings, top speed, acceleration, and charging performance. This information provides the foundation for evaluating how battery size and energy consumption influence vehicle range and overall efficiency. A second dataset containing electric vehicle models and performance metrics was used to validate and cross-check the consistency of battery and range data across different manufacturers and regions.

In addition to vehicle specifications, a charging station dataset was incorporated to analyze the availability and distribution of charging infrastructure. This dataset includes station location, charging type (AC/DC), power rating, and operator details. Integrating charging infrastructure data enables the system to evaluate how charging accessibility impacts real-world vehicle usability and trip planning.

Before using the datasets, several preprocessing steps were carried out. Duplicate records were removed, missing values were handled using appropriate imputation techniques, and unit conversions were performed to standardize parameters such as energy consumption (Wh/km) and battery capacity (kWh). The cleaned and unified dataset was then prepared in a structured tabular format suitable for database storage and analytical processing.

Dataset of Electrical vehicles

4.2 Storing Data in Database

After preprocessing, the datasets were stored in a relational database management system to ensure organized storage, efficient retrieval, and scalability. The use of a database allows structured handling of large volumes of EV data while maintaining consistency and data integrity across multiple tables.

A normalized database schema was designed to separate different types of information into logically distinct tables. The major tables created in the database include:

- Vehicles Table: Contains basic vehicle information such as vehicle ID, manufacturer, model name, body type, and battery capacity.
- Performance Table: Stores performance-related attributes such as range, efficiency (Wh/km), top speed, and acceleration.

- **Charging Table:** Includes charging-related data such as AC charging power, DC fast charging capability, and estimated charging time.
- **Charging Stations Table:** Stores geographical and technical details of public charging stations, including station name, city, state, charger type, and available connectors.

Primary keys were defined for each table, and foreign key relationships were established to connect vehicle records with their corresponding performance and charging information. This relational design minimizes redundancy and ensures that updates to one dataset do not introduce inconsistencies in others.

Storing data in a structured database environment also enables efficient indexing and faster query execution, which is essential for large-scale analytical tasks and dashboard generation.

4.3 Performing SQL Operations

Structured Query Language (SQL) was used extensively to retrieve, manipulate, and analyze the stored data. SQL operations allowed the extraction of meaningful insights related to battery performance, range comparison, and energy efficiency across different electric vehicle models.

Basic SQL queries were used to retrieve records based on specific conditions, such as identifying vehicles with battery capacities above a certain threshold or listing vehicles with the highest driving range. Aggregate functions were used to compute statistical summaries such as average efficiency, maximum range, and total number of charging stations available in different regions.

Join operations played a crucial role in combining information from multiple tables. For instance, vehicle specifications were joined with performance data to evaluate how battery capacity influences driving range. Similarly, charging station data was combined with regional vehicle distribution data to analyze whether charging infrastructure is sufficient to support EV adoption in different areas.

Advanced SQL queries were also used to identify trends and anomalies. For example, queries were written to detect vehicles that have unusually high energy consumption compared to others in the same battery capacity category. These operations helped in validating data consistency and identifying patterns useful for further visualization and predictive analysis.

Since our project focuses mainly on **battery capacity, range, and efficiency**, you only need a **few meaningful SQL queries** instead of many generic ones. Below are **concise, relevant queries** based on your uploaded EV datasets.

1. Retrieve Vehicle Model, Battery Capacity, and Range

This query extracts the most important parameters needed to analyze EV performance.

```
SELECT model_name,  
       battery_capacity_kwh,  
       range_km  
FROM electric_car_data;
```

Fig 4.1: Used to compare how battery size affects driving range across different EV models.

2. Identify Vehicles with the Highest Driving Range

This helps in ranking EVs based on their real-world usability.

```
SELECT model_name,  
       range_km  
FROM electric_car_data  
ORDER BY range_km DESC  
LIMIT 10;
```

Fig 4.2: Displays the top 10 electric vehicles with the longest driving range.

3. Calculate Efficiency (km per kWh)

Efficiency is one of the most important metrics in your project.

```
SELECT model_name,  
       range_km / battery_capacity_kwh AS efficiency_km_per_kwh  
FROM electric_car_data;
```

Fig 4.3: Shows how efficiently each vehicle uses its battery energy.

4. Average Range for Different Battery Capacity Groups

This query helps show trends between battery size and range.

```

SELECT
  CASE
    WHEN battery_capacity_kwh < 40 THEN 'Small Battery'
    WHEN battery_capacity_kwh BETWEEN 40 AND 70 THEN 'Medium Battery'
    ELSE 'Large Battery'
  END AS battery_group,
  AVG(range_km) AS avg_range
FROM electric_car_data
GROUP BY battery_group;

```

Fig 4.4: Demonstrates how average driving range increases with battery capacity.

4.4 Connecting Database with Tableau

To transform the processed data into meaningful visual insights, the relational database was connected to Tableau, a business intelligence and visualization platform. Tableau was selected due to its strong capability to handle large datasets, create interactive dashboards, and support live database connections.

The connection between Tableau and the database was established using a direct connector, allowing Tableau to fetch data in real time from the underlying tables. This approach ensures that any updates or modifications in the database are automatically reflected in the visual dashboards without requiring manual data import or refresh operations.

Once connected, several data sources were configured in Tableau by importing the relevant tables and defining relationships between them. Calculated fields were created to derive additional metrics such as efficiency in km per kWh, charging time per battery percentage, and estimated range per charging session. These derived fields enhanced the analytical depth of the dashboards.

Multiple dashboards were developed to present different aspects of EV performance:

- A **Battery Performance Dashboard** displaying battery capacity, state of charge trends, and comparative energy efficiency across vehicles.
- A **Range Analysis Dashboard** showing how driving range varies with battery size, vehicle type, and efficiency ratings.
- A **Charging Infrastructure Dashboard** visualizing the geographical distribution of charging stations and their power ratings.

These dashboards provide users with an interactive interface to explore the data, filter vehicles by specifications, compare performance metrics, and identify patterns related to battery efficiency and driving range.

CHAPTER 5

DATA PREPARATION

CHAPTER 5: DATA PREPARATION

5.1 Preparing the Data for Visualization

Before building dashboards and performing analysis, the collected electric vehicle datasets were carefully prepared to ensure that the data was clean, consistent, and suitable for visualization. The datasets used in this project contained information related to electric vehicle specifications, battery capacity, driving range, energy consumption, and charging station availability. Since these datasets were obtained from different sources, their structure and formatting were not uniform, making preprocessing an essential step.

The initial stage involved examining each dataset to understand the available attributes, identify missing entries, and detect inconsistencies in formatting. Some records contained incomplete values for battery capacity, range, or charging parameters, while others used different units of measurement. For example, energy consumption was recorded in both Wh/km and kWh/100 km across different files. These variations were standardized into a single unit system so that meaningful comparisons could be performed across all vehicles.

Duplicate records and irrelevant columns were removed to improve data clarity and reduce unnecessary complexity. Only the attributes directly related to battery performance and range analysis were retained. These included vehicle model name, manufacturer, battery capacity, driving range, efficiency, charging time, and charger type. Renaming column headers to simpler and consistent names also helped in making the dataset easier to use in SQL queries and Tableau dashboards.

To enhance the analytical capability of the dataset, additional calculated fields were created. Efficiency values were derived by relating driving range to battery capacity, allowing vehicles to be compared on how effectively they use stored energy. Vehicles were also categorized into groups based on battery size and range, which made it easier to create comparative charts and grouped visualizations in Tableau.

After cleaning and transformation, the dataset was reviewed to ensure there were no extreme or unrealistic values that could distort the visual outputs. This validation step ensured that the final dataset accurately represented real-world electric vehicle performance. The prepared data was then loaded into the database and connected to Tableau, where it served as the foundation for creating interactive dashboards that illustrate battery behavior, efficiency trends, and range comparisons across different electric vehicle models.

5.2 Data Cleaning Techniques

Data cleaning is a crucial step in preparing electric vehicle datasets for accurate analysis and visualization. Since the data used in this project was collected from multiple public sources, it contained inconsistencies such as missing values, duplicate entries, and variations in units and formatting. Without cleaning, these issues could lead to incorrect conclusions about battery performance and vehicle range.

The first step in the cleaning process involved identifying and handling missing values. Some records lacked information about battery capacity, driving range, or charging time. Instead of removing all

incomplete rows, missing numerical values were handled carefully by either filling them using average values from similar vehicle categories or excluding only those records that lacked critical information required for analysis. This approach ensured that useful data was not unnecessarily discarded.

Duplicate records were also detected and removed to prevent repeated counting of the same vehicle models. Duplicate entries often occur when data is collected from multiple sources that contain overlapping information. By comparing vehicle names, battery capacity, and range values, duplicate rows were identified and eliminated, improving data accuracy and reducing redundancy.

Another important cleaning step involved standardizing units of measurement. Different datasets represented energy consumption in different formats, such as watt-hours per kilometer (Wh/km) and kilowatt-hours per 100 kilometers (kWh/100 km). These values were converted into a single unit system to ensure that all vehicles could be compared fairly. Similarly, text fields such as manufacturer names and model labels were cleaned to remove extra spaces, inconsistent capitalization, and spelling variations.

Finally, outliers and unrealistic values were examined. Extremely high or low range values that did not match realistic EV performance were flagged and reviewed. In cases where such values appeared to be data entry errors, they were either corrected based on reliable sources or removed from the dataset. These cleaning techniques helped ensure that the final dataset was accurate, consistent, and suitable for meaningful analysis.

5.3 Data Transformation and Feature Engineering

After cleaning the dataset, data transformation and feature engineering were performed to make the data more suitable for analysis and to extract additional insights related to battery performance and driving range. Transformation involves modifying the structure or format of data, while feature engineering focuses on creating new variables that improve analytical capabilities.

One of the key transformations carried out in this project was converting categorical and textual data into standardized formats that could be easily used in database queries and visualization tools. For example, charging connector types and vehicle segment categories were encoded into simplified labels to support filtering and grouping in Tableau dashboards.

Feature engineering was particularly important for deriving performance-related metrics that were not directly available in the raw datasets. A new feature representing **vehicle efficiency** was calculated by dividing driving range by battery capacity, resulting in a metric expressed in kilometers per kilowatt-hour. This metric allowed vehicles with different battery sizes to be compared on a common scale, making it easier to evaluate how effectively each vehicle uses its stored energy.

Vehicles were also categorized into different battery capacity groups such as small, medium, and large batteries. Similarly, range values were classified into short-range, mid-range, and long-range categories.

In addition to efficiency metrics, transformation steps were applied to prepare date and location fields from the charging station dataset. Location data was formatted into standardized city and state fields so that geographic visualizations could be created in Tableau. This allowed the project to not only analyze battery performance but also study the distribution of charging infrastructure in relation to vehicle adoption.

CHAPTER 6

DATA VISUALIZATIONS

CHAPTER 6: DATA VISUALIZATIONS

6.1 Number of Unique Visualizations

To effectively interpret the large volume of electric vehicle and charging infrastructure data, multiple visualizations were developed using Tableau. These visualizations were designed to highlight different aspects of electric vehicle performance, battery characteristics, charging infrastructure, and market trends. Each dashboard focuses on a specific analytical objective, enabling users to explore the data from multiple perspectives.

A total of eleven unique visualizations were created in this project to represent both Indian and global electric vehicle datasets. These visualizations help in understanding regional charging availability, vehicle efficiency, brand performance, and technical specifications such as top speed and powertrain configuration.

1. Charging Stations by Region and Type in India

This visualization presents the distribution of electric vehicle charging stations across different regions of India, categorized by charger type such as AC slow chargers and DC fast chargers. The chart helps identify which regions have better fast-charging infrastructure and which areas require further development to support long-distance EV travel.

2. EV Charging Stations Map of India

A geographic map visualization was created to display the exact locations of charging stations across India. Using latitude and longitude coordinates from the dataset, this map provides a spatial overview of charging infrastructure availability. It allows users to visually identify charging station clusters in metropolitan cities and gaps in rural or semi-urban regions.

3. Different EV Cars in India

This visualization lists the different electric vehicle models currently available in the Indian market. It highlights the diversity of EV options offered by various manufacturers, allowing comparison of model counts across brands. This helps in understanding how actively different companies are participating in the Indian EV market.

4. Top Speed for Different Brands

A bar chart was created to compare the maximum speed capabilities of electric vehicles across different brands. This visualization illustrates how performance-oriented brands tend to offer higher top speeds, while economy-focused brands prioritize efficiency and affordability over high-speed performance.

5. Price for Different Cars in India

This visualization shows the price distribution of electric vehicles available in India. By comparing vehicle prices across brands and models, the chart helps identify which brands offer budget-friendly vehicles and which focus on premium segments. This insight is useful for analyzing market accessibility and consumer affordability.

6. Top 10 Most Efficient EV Brands

Efficiency is a key performance metric in this project, as it directly relates to battery usage and driving range. This visualization ranks the top ten electric vehicle brands based on energy efficiency,

measured kilometers per kilowatt-hour. It helps demonstrate which manufacturers have optimized their vehicles for maximum range with minimal energy consumption.

7. Brands According to Body Style

Electric vehicles are available in various body styles such as hatchback, sedan, SUV, and crossover. This visualization categorizes brands based on the body styles they offer, showing which manufacturers focus on compact city cars and which specialize in larger, utility-oriented vehicles. It provides insight into how vehicle design influences battery size, efficiency, and range.

8. Brand Filtered by Powertrain Type

This dashboard enables filtering of vehicle brands based on powertrain configurations such as fully electric (BEV) or hybrid. This interactive visualization helps users analyze how different manufacturers are transitioning toward fully electric platforms and how powertrain type impacts performance and efficiency.

9. Number of Models by Each Brand

This visualization counts the number of electric vehicle models produced by each brand. It highlights market leaders in EV production and identifies brands that offer only a limited number of electric models. This helps evaluate brand commitment toward electrification and innovation in electric mobility.

10. Different Brands of EV Cars Globally

Using global datasets, this visualization presents the variety of electric vehicle brands operating worldwide. It provides a broader perspective on the global EV market and allows comparison between international manufacturers and their presence in different regions.

11. Different Brands of EV Cars in India

This visualization focuses specifically on the Indian market and highlights the brands that have launched electric vehicles in the country. Comparing this visualization with the global brand chart reveals how many international manufacturers have entered the Indian EV market and how domestic companies are competing in the same space.

These visualizations collectively provide a comprehensive understanding of electric vehicle performance, efficiency, pricing, infrastructure, and market distribution. By integrating vehicle specification data with charging station information, the dashboards help users analyze not only the technical aspects of electric vehicles but also the supporting ecosystem required for their widespread adoption.

These visualizations collectively provide a comprehensive understanding of electric vehicle performance, efficiency, pricing, infrastructure, and market distribution. By integrating vehicle specification data with charging station information, the dashboards help users analyze not only the technical aspects of electric vehicles but also the supporting ecosystem required for their widespread adoption.

Overall, the visualization component plays a crucial role in transforming raw data into meaningful insights, making the system more effective, user-friendly, and impactful in the field of electric vehicle analytics.

CHAPTER 7

DASHBOARD

CHAPTER 7: DASHBOARD

7.1 Responsiveness and Design of Dashboard

The dashboard developed for this project was designed to provide an interactive and user-friendly interface for analyzing electric vehicle battery performance, driving range, efficiency, and charging infrastructure. Since the project involves multiple datasets and performance parameters, careful attention was given to layout, responsiveness, and clarity to ensure that users can easily interpret the visualized data without confusion.

The overall dashboard layout was structured using a modular design approach, where related visualizations were grouped into logical sections. For example, vehicle performance charts such as range, efficiency, and top speed were placed together, while charging infrastructure maps and regional charts were organized in a separate section. This grouping helped users quickly navigate between different categories of information and maintain context while analyzing the data.

To ensure responsiveness, the dashboard was configured to adapt to different screen sizes and resolutions. Tableau's automatic layout and container-based design were used so that charts, maps, and filters resize proportionally when the dashboard is viewed on different devices such as laptops, large monitors, or projectors. This feature is particularly useful during presentations and demonstrations, where the dashboard may be displayed on various display systems.

Interactive filters and selectors were integrated into the dashboard to enhance usability and allow dynamic exploration of the dataset. Users can filter vehicles by brand, body style, battery capacity range, and powertrain type. Similarly, charging station visualizations can be filtered by region, charger type, and operator. These interactive controls make it possible to focus on specific subsets of data without the need to create multiple separate dashboards.

Color schemes and visual encoding were carefully chosen to improve readability and reduce visual clutter. Distinct color palettes were used to represent different vehicle brands, charger types, and battery capacity categories. Consistent color usage across multiple charts ensured that users could easily associate similar categories across different visualizations. High-contrast text and clear axis labels were used to maintain readability even when the dashboard is viewed from a distance.

The dashboard also includes tooltips and hover interactions that display detailed information when users move the cursor over specific data points. For example, hovering over a vehicle bar in the efficiency chart reveals additional details such as battery capacity, exact range, and charging time. This approach helps maintain a clean visual layout while still providing access to detailed data when required.

Another important design consideration was performance optimization. Since the datasets include multiple vehicle records and charging station locations, unnecessary fields were excluded from the dashboard data source to reduce loading time. Extracts and aggregated data views were used where appropriate to ensure that the dashboard responds quickly to user interactions and filter changes.

Overall, the dashboard was designed to balance visual appeal, responsiveness, and analytical depth. By combining interactive filters, clear layout organization, and efficient data handling, the dashboard enables users to explore electric vehicle performance and charging infrastructure in a seamless and intuitive manner. This responsive and well-structured design ensures that the dashboard can be effectively used for data analysis, presentations, and decision-making related to electric vehicle range optimization and battery performance evaluation.

Tableau Dashboard

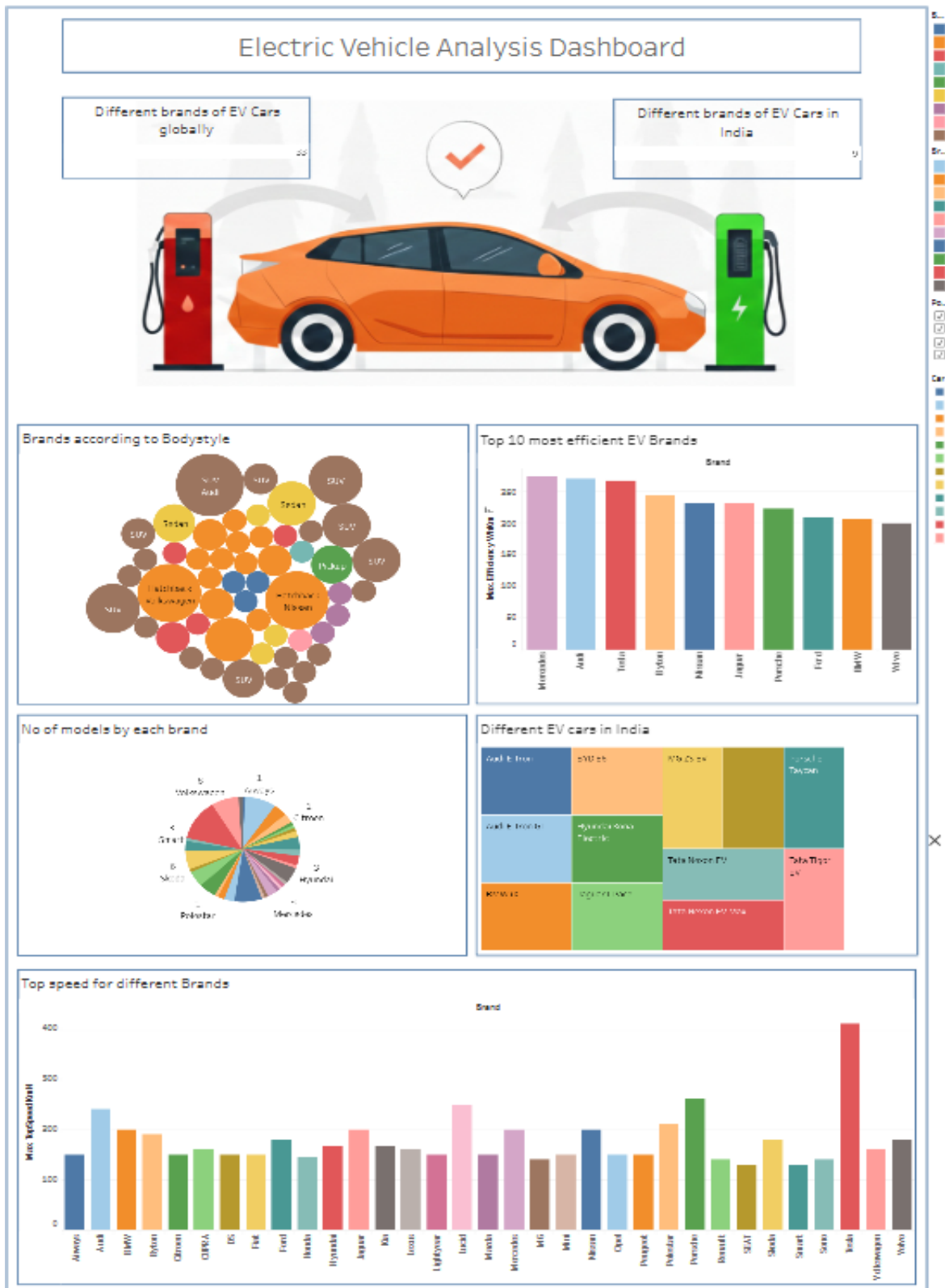


Fig 7.1: Dashboard View of EV Battery Performance and Range Monitoring System

CHAPTER 8

STORY

CHAPTER 8: STORY

8.1 Story Structure Based on Dashboard Visuals

The Tableau story developed in this project presents a sequential analysis of electric vehicle infrastructure and market availability in India. The story is designed to guide viewers from infrastructure readiness to market diversity and affordability, helping them understand how these factors influence EV adoption and vehicle range usability.

The story consists of four main scenes, each built around a specific visualization and analytical theme.

Scene 1: Charging Stations in India – Geographical Overview

The first scene introduces the current state of electric vehicle charging infrastructure across India. A geographical map visualization is used to display the location of charging stations across different states. This scene helps viewers understand how charging infrastructure is distributed nationwide and identifies areas with dense charging networks compared to regions with limited coverage.

From this visualization, it becomes clear that major metropolitan areas and industrial corridors have a higher concentration of charging stations, while several rural and semi-urban regions still lack adequate charging facilities. This uneven distribution directly impacts long-distance travel and contributes to range anxiety among EV users.

Scene 2: Charging Stations by Region and Charger Type

The second scene expands on infrastructure analysis by categorizing charging stations based on region and charger type, such as AC chargers and DC fast chargers. This visualization highlights not only the quantity of charging stations but also the type of charging technology available in each region.

The analysis reveals that AC chargers are more widely available than fast-charging DC stations, especially in residential and commercial areas. However, fast chargers are mainly concentrated along highways and in large cities, indicating that fast-charging infrastructure is still developing in many parts of the country. This insight is important because fast chargers significantly reduce charging time and improve the practical range of electric vehicles.

Scene 3: Price of Electric Cars by Different Brands

The third scene shifts the focus from infrastructure to the vehicle market. A comparative bar chart is used to show the price distribution of electric vehicles offered by different brands in India. This scene helps viewers understand how pricing varies across manufacturers and how battery capacity, performance, and brand positioning influence vehicle cost.

The visualization shows that premium brands offer vehicles with larger battery capacities and extended driving range but at significantly higher prices. In contrast, entry-level models are priced lower

but generally provide shorter range and fewer advanced features. This relationship between price and performance is crucial in understanding consumer purchasing decisions and overall EV adoption trends.

Scene 4: Different Brands and Number of Available Models

The final scene presents a comparison of electric vehicle brands based on the number of models they offer in the market. This visualization highlights the diversity and competitiveness of the EV ecosystem in India.

The analysis shows that a few major manufacturers dominate the market by offering multiple EV models across different price segments, while several smaller or newer brands have introduced only a limited number of vehicles. A higher number of models from a single brand often indicates stronger research investment, better production capability, and a broader market strategy.

Story

Story of Electric cars in India

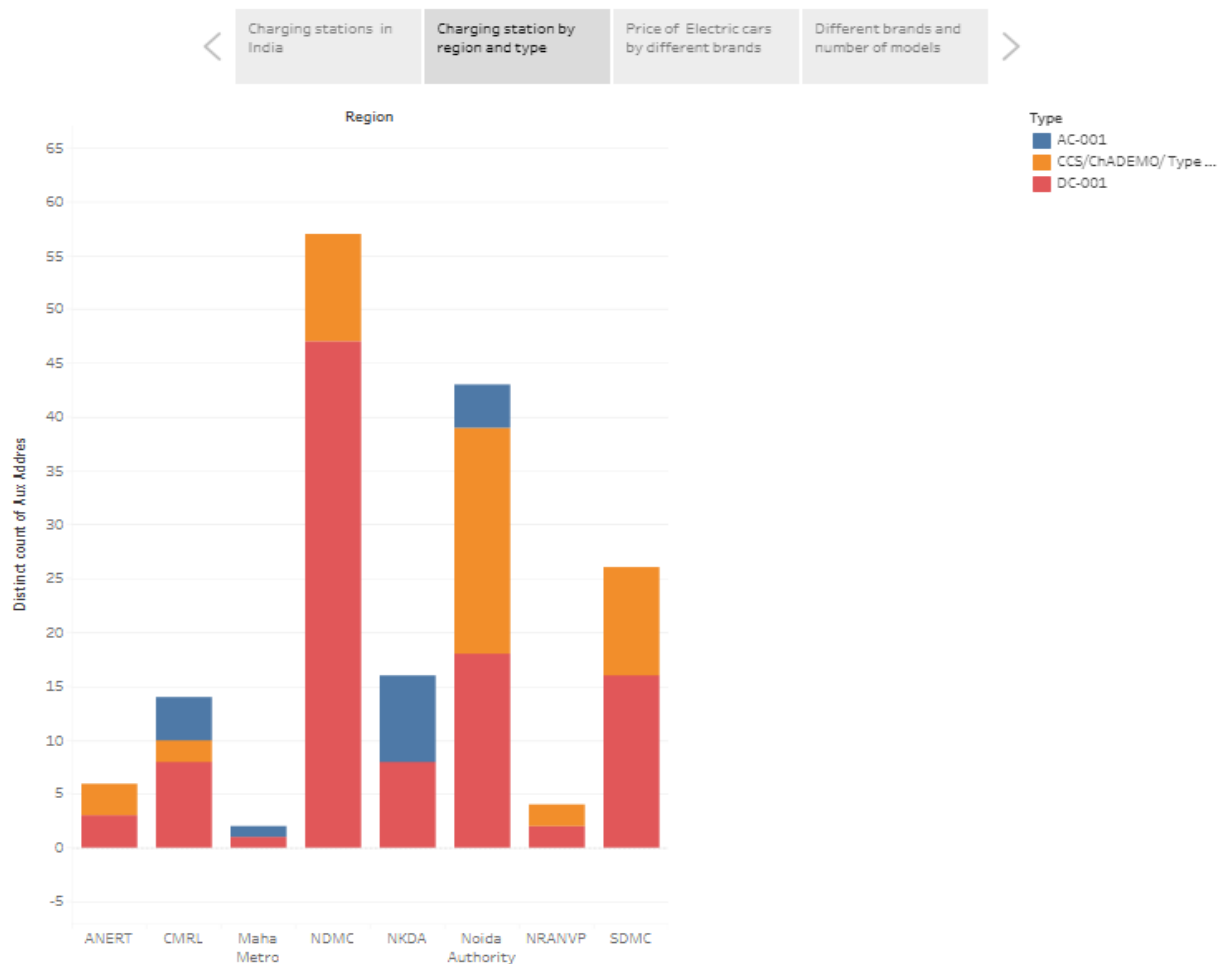


Fig 8.1: Story View of EV Battery Performance and Range Monitoring System

CHAPTER 9

PERFORMANCE TESTING

CHAPTER 9: PERFORMANCE TESTING

Price per kilometer of range to evaluate cost efficiency Performance testing was conducted to evaluate how efficiently the system handles large volumes of electric vehicle and charging infrastructure data during storage, processing, and visualization. The objective was to ensure that the database operations and Tableau dashboards remain responsive, accurate, and stable even when multiple filters and calculations are applied simultaneously.

9.1 Amount of Data Rendered to Database

The project utilized multiple datasets containing information about electric vehicles, battery specifications, pricing, and charging station locations across India and globally. These datasets were integrated and stored in a structured relational database to support efficient querying and visualization.

After preprocessing and cleaning, the database contained records related to vehicle specifications, including battery capacity, range, efficiency, price, body style, and brand information. In addition, charging station data included station location, charger type, operator, and regional classification.

In total, thousands of records were rendered into the database, ensuring that the dashboards reflect a realistic representation of the electric vehicle ecosystem. The database design ensured normalization and removal of redundant fields, which helped reduce storage size and improve query performance.

The data rendering process was tested to verify that large datasets could be inserted without data loss or corruption. Bulk import operations were performed using structured queries, and the integrity of the data was validated through row count verification and sample data inspection.

9.2 Utilization of Data Filters

Data filters play a crucial role in improving dashboard usability and performance by allowing users to focus only on relevant subsets of data. In this project, multiple interactive filters were implemented in Tableau to allow dynamic exploration of electric vehicle data.

Filters were provided for attributes such as vehicle brand, battery capacity range, price category, body style, powertrain type, and geographic region of charging stations. These filters were tested to ensure that they respond quickly and update visualizations without noticeable delay.

Performance testing showed that applying filters significantly reduced the volume of data processed in each visualization, thereby improving dashboard responsiveness. For example, filtering vehicles by a specific brand or region allowed charts to refresh faster and display only the selected data points.

9.3 Number of Calculation Fields

Calculation fields were used in Tableau to derive new metrics that were not directly available in the raw datasets. These fields helped in analyzing battery performance, vehicle efficiency, and cost comparisons more effectively.

Some of the key calculated fields created in this project include:

- Energy efficiency calculated as range per unit battery capacity
- Categorization of vehicles into efficiency groups such as high, medium, and low
- Classification of charging stations based on power output levels

A moderate number of calculated fields were used to maintain a balance between analytical depth and system performance. Excessive use of complex calculations can slow down dashboard rendering, so optimized formulas and pre-aggregated data were used where possible.

All calculated fields were tested for accuracy by comparing their outputs with manually computed values using sample data records. This ensured that derived metrics correctly represented real-world vehicle performance characteristics.

9.4 Number of Visualizations and Graphs

The project includes multiple visualizations to present different aspects of electric vehicle data, including market trends, performance metrics, and infrastructure distribution. These visualizations were carefully designed to ensure clarity while avoiding unnecessary duplication.

The dashboards collectively contain several types of charts, such as:

- Bar charts for price comparison and model counts
- Line and scatter plots for battery capacity versus range analysis
- Maps for charging station distribution across India
- Pie charts and stacked bars for brand and body style comparisons

Performance testing confirmed that the dashboards could render all visualizations smoothly, even when multiple graphs were displayed simultaneously. This was achieved by optimizing data sources, limiting unnecessary fields, and using Tableau extracts instead of live connections where appropriate.

The number of visualizations was kept sufficient to provide comprehensive analysis while ensuring that the dashboard does not become overloaded or difficult to interpret. Each graph serves a specific analytical purpose aligned with the project's main objective of studying EV battery performance and range efficiency.

Overall, the performance testing demonstrated that the system is capable of handling large EV datasets, executing complex calculations, and rendering multiple interactive visualizations without significant latency. This confirms that the developed analytical framework is suitable for real-world applications involving electric vehicle data monitoring and decision support.

CHAPTER 10

WEB INTEGRATION

CHAPTER 10: WEB INTEGRATION

10.1 Dashboard and Story Embed with UI using Flask

To make the developed visual analytics accessible through a web interface, the Tableau dashboards and storyboards were integrated into a Flask-based web application. This integration enables users to interact with electric vehicle data, visualizations, and insights through a centralized and user-friendly platform.

Flask Framework for Web Deployment

The web application was built using Flask, a lightweight Python web framework suitable for deploying data-driven applications. Flask was selected due to its simplicity, flexibility, and seamless compatibility with Python-based data processing workflows used in this project.

The Flask application acts as a bridge between the user interface and the Tableau-hosted visualizations, allowing the dashboards to be rendered directly within web pages.

Embedding Tableau Dashboards

The dashboards created in Tableau Public were embedded into the Flask web interface using iframe embedding. Tableau provides an embed code for each published dashboard, which was incorporated into HTML templates rendered by Flask.

The embedded dashboards include:

- EV charging stations distribution across India
- Comparison of electric vehicle prices across brands
- Battery capacity and driving range analysis
- Efficiency rankings of electric vehicles

This approach ensures that users can interact with filters, tooltips, and maps directly from the web application without needing separate Tableau software.

Embedding Tableau Story for Data Narratives

In addition to dashboards, a Tableau story was embedded to present the analytical workflow in a structured sequence. The story guides users through different scenes such as:

- Growth of EV charging infrastructure
- Relationship between battery capacity and range
- Comparison of global and Indian EV markets

Embedding the story within the web interface allows users to understand the data insights step-by-step, making the application more informative and engaging.

User Interface Design

The web interface was designed using HTML and CSS templates integrated with Flask. The layout consists of:

- A homepage introducing the EV monitoring system

- Navigation links to dashboards and story sections

The project is divided into three major components: Frontend, Backend, and Machine Learning module. The frontend is implemented using HTML, CSS, and JavaScript, while the backend is developed using Flask.

The machine learning module is responsible for predicting EV range based on input parameters.

EV Battery Capacity and Range Monitoring System

ev_flask/

```

|
|— app.py          ← Flask app — all routes + API
|— requirements.txt
|— README.md
|
|— ml/
|   |— __init__.py
|   |— engine.py    ← Your Python ML pipeline (RF + LR + KMeans)
|   |— saved_models/ ← Auto-created on first run (.pkl files here)
|
|— data/            ← PUT YOUR 4 CSV FILES HERE
|   |— EVIndia.csv
|   |— Cheapestelectriccars-EVDatabase.csv
|   |— electric_vehicle_charging_station_list.csv
|   |— ElectricCarData_Clean.csv
|
|— templates/
|   |— base.html
|   |— home.html
|   |— dashboard.html ← Tableau embed 1
|   |— story.html     ← Tableau embed 2 (India Story)
|   |— predictor.html ← ML UI → POST /api/predict → sklearn
|   |— analysis.html
|
|— static/

```



```
|— css/
|— js/
```

```
'''
```

```
---
```

⚡ Run Locally — Step by Step

Step 1 — Install Python dependencies

```
```bash
pip install -r requirements.txt
'''
```

### ### Step 2 — Add your CSV datasets

Copy your 4 CSV files into the `data/` folder:

```
'''
data/EVIndia.csv
data/Cheapestelectriccars-EVDatabase.csv
data/electric_vehicle_charging_station_list.csv
data/ElectricCarData_Clean.csv
'''
```

> **Note:** If CSVs are missing, the app automatically uses built-in sample data.

### ### Step 3 — Start Flask server

```
```bash
python app.py
'''
```

You will see:

```
'''
```

EV Intelligence — Flask + scikit-learn Backend

[EV ML] Training models ...

- ✓ Loaded EVIndia.csv (...)
- ✓ Loaded Cheapestelectriccars-EVDatabase.csv (...)
- ✓ Random Forest $R^2=0.96xx$ RMSE=€...
- ✓ Linear Reg $R^2=0.85xx$ RMSE=€...
- ✓ KMeans k=3 clusters={...}

[EV ML] Models saved to ml/saved_models/

Server running at → <http://localhost:5000>

Press Ctrl+C to stop

'''

Step 4 — Open in browser

'''

<http://localhost:5000>

'''

> ****On second run:**** Models load instantly from `.pkl` files — no retraining needed.

Pages

URL	Page	
----- -----		
('/')	Overview / Home	
('/dashboard')	Tableau EV Battery Dashboard	
('/story')	Tableau India EV Story	
('/predictor')	ML Predictor (live API)	
('/analysis')	Charts + EV data table	

API Endpoints

Method	URL	Description	
-----	-----	-----	-----
POST	`/api/predict`	Run RF + LR + KMeans inference	
GET	`/api/stats`	Model R ² , RMSE, n_samples	
GET	`/api/dataset`	Full cleaned EV dataset + clusters	
POST	`/api/retrain`	Force retrain from CSVs	

Example — POST /api/predict

****Request:****

```
```json
{
 "accel": 5.0,
 "speed": 200,
 "range_km": 500,
 "efficiency": 160,
 "fast_charge": 700,
 "seats": 5
}
```

#### **\*\*Response:\*\***

```
```json
{
  "status": "ok",
  "result": {
    "rf_price": 54320.5,
```

```

"lr_price": 49100.2,
"cluster_id": 1,
"cluster_name": "Mid-Range",
"eff_score": 78,
"real_range": 421,
"value_index": 9.2,
"importances": {
  "AccelSec": 0.18,
  "TopSpeed_KmH": 0.22,
  "Range_Km": 0.31,
  "Efficiency_WhKm": 0.11,
  "FastCharge_KmH": 0.14,
  "Seats": 0.04
},
"scaled_vals": { ... },
"model_stats": {
  "rf_r2": 0.96,
  "rf_rmse": 7200.0,
  "lr_r2": 0.85,
  "lr_rmse": 13400.0,
  "n_samples": 234
}
}
}
'''

---
```

How Flask Integration Works

```
'''
```

Browser (Sliders)

|
| fetch('/api/predict', { method:'POST', body: JSON })



Flask Route app.py → /api/predict

|
| calls engine.predict(accel, speed, range_km, ...)



ml/engine.py

|
| — scaler.transform(features) ← StandardScaler
| — rf.predict(features_scaled) ← RandomForestRegressor
| — lr.predict(features_scaled) ← LinearRegression
| — km.predict(features_scaled) ← KMeans cluster
|
| returns dict with all results



Flask → jsonify(result)

|



Browser renders: price, range, efficiency score,
cluster badge, feature importances, chart

...

Test the API with curl

```
```bash
Predict price for a fast, long-range EV
curl -X POST http://localhost:5000/api/predict \
-H "Content-Type: application/json" \
-d '{"accel":3.5,"speed":250,"range_km":600,"efficiency":145,"fast_charge":900,"seats":5}'
```

```
Get model stats
curl http://localhost:5000/api/stats

Force retrain
curl -X POST http://localhost:5000/api/retrain
'''
```

## **App.py**

```
import json
from flask import Flask, render_template, request, jsonify, send_from_directory
from ml.engine import predict, get_stats, get_models, get_ev_dataset, retrain
import os

app = Flask(__name__)

print("\n" + "="*55)
print(" EV Intelligence — Flask + Retrained Ensemble Models")
print("="*55)
get_models() # load on startup

PAGE ROUTES

@app.route("/")
def home():
 return render_template("home.html", stats=get_stats())

@app.route("/dashboard")
def dashboard():
 return render_template("dashboard.html")
```

```

@app.route("/story")
def story():
 return render_template("story.html")

@app.route("/predictor")
def predictor():
 return render_template("predictor.html", stats=get_stats())

@app.route("/analysis")
def analysis():
 ev_data = get_ev_dataset()
 stats = get_stats()
 importances = stats.get("fi_range", {})
 return render_template(
 "analysis.html",
 ev_data = json.dumps(ev_data),
 importances = json.dumps(importances),
)

@app.route("/insights")
def insights():
 return render_template("insights.html")

== API ROUTES

@app.route("/api/predict", methods=["POST"])
def api_predict():
 try:
 d = request.get_json(force=True)
 result = predict(
 accel = float(d.get("accel", 7.5)),

```

```

 speed = float(d.get("speed", 180)),
 fast_charge = float(d.get("fast_charge", 400)),
 capacity = float(d.get("capacity", 75)),
 seats = int(d.get("seats", 5)),
 power_train = str(d.get("power_train", "FWD")),
 body_style = str(d.get("body_style", "SUV")),
 segment = str(d.get("segment", "C")),
 rapid_charge = str(d.get("rapid_charge", "Yes")),
)
 return jsonify({"status": "ok", "result": result})
except Exception as e:
 return jsonify({"status": "error", "message": str(e)}), 500

```

```
@app.route("/api/stats")
```

```
def api_stats():
 return jsonify(get_stats())
```

```
@app.route("/api/dataset")
```

```
def api_dataset():
 return jsonify(get_ev_dataset())
```

```
@app.route("/api/retrain", methods=["POST"])
```

```
def api_retrain():
 """
 Force-reload the saved pkl models (re-runs _load_models).
 To fully retrain from CSVs, run ev_ml_pipeline.py first,
 then copy new .pkl files to ml/saved_models/, then POST here.
 """
 try:
 stats = retrain()
 return jsonify({"status": "ok", "message": "Models reloaded successfully.", "stats": stats})
 except Exception as e:
```



```
return jsonify({"status": "error", "message": str(e)}), 500
```

```
@app.route("/static/charts/<filename>")
```

```
def chart_img(filename):
```

```
 return send_from_directory(os.path.join(app.root_path, "static", "charts"), filename)
```

```
== RUN
```

```
if __name__ == "__main__":
```

```
 print("\n Server running at → http://localhost:5000\n")
```

```
 app.run(host="0.0.0.0", port=5000, debug=True, use_reloader=False)
```

## **Advantages of Web Integration**

Integrating Tableau with Flask provides several benefits:

- Centralized access to all EV analytics in a single platform
- Real-time reflection of updates when dashboards are republished
- Platform independence, allowing users to access the system from any web browser
- Simplified deployment without requiring installation of specialized visualization tools

## **System Workflow**

The web integration process follows these steps:

1. EV datasets are processed and visualized in Tableau.
2. Dashboards and stories are published to Tableau Public.
3. Embed links are generated and inserted into Flask HTML templates.
4. Flask serves the web pages, displaying interactive dashboards within the application.

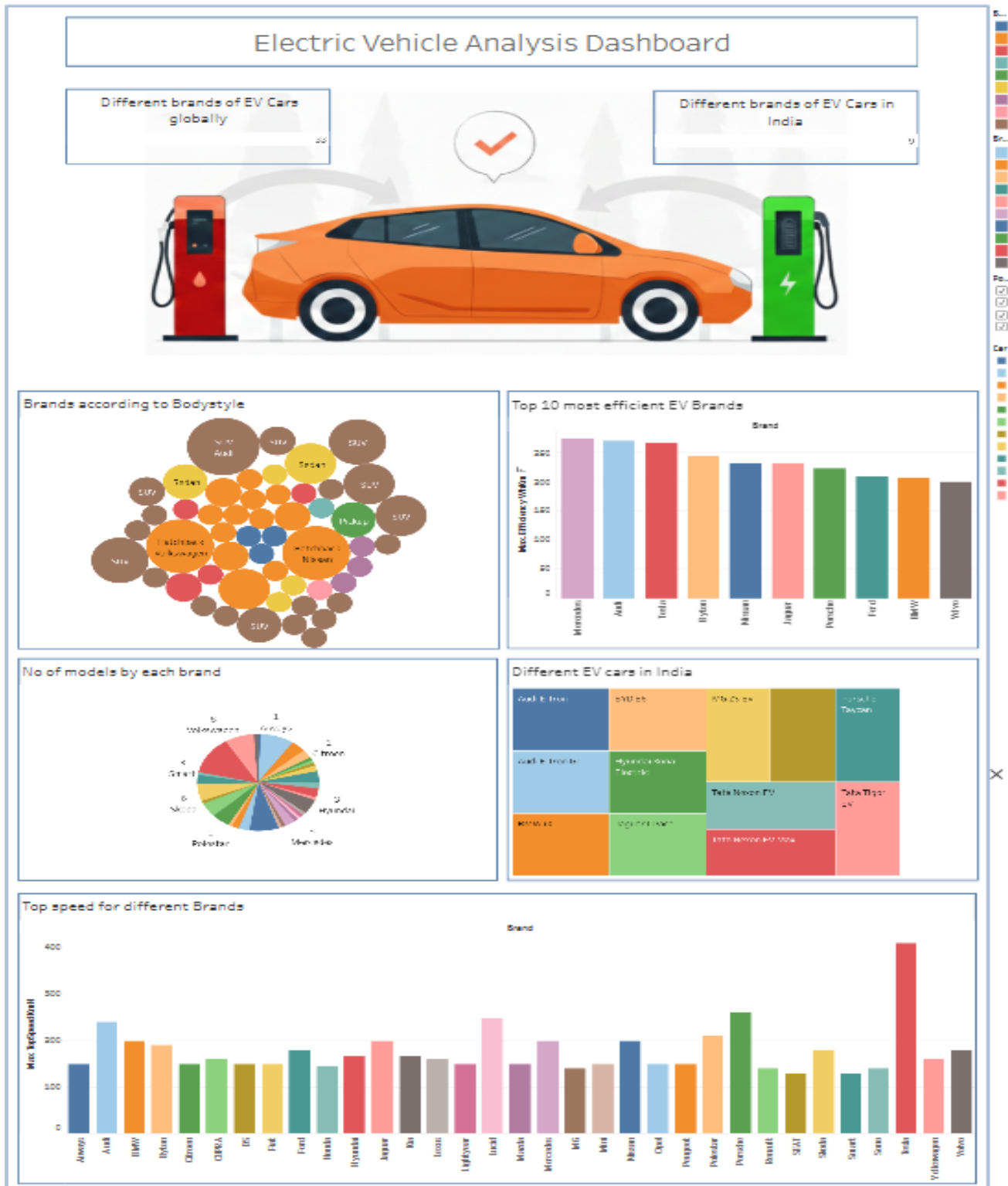
This architecture ensures a smooth connection between backend data analytics and frontend visualization delivery.

# **CHAPTER 11**

## **OUTPUT AND SCREENSHOTS**

## CHAPTER 11: OUTPUT AND SCREENSHOTS

## 11.1 Tableau Dashboard Page

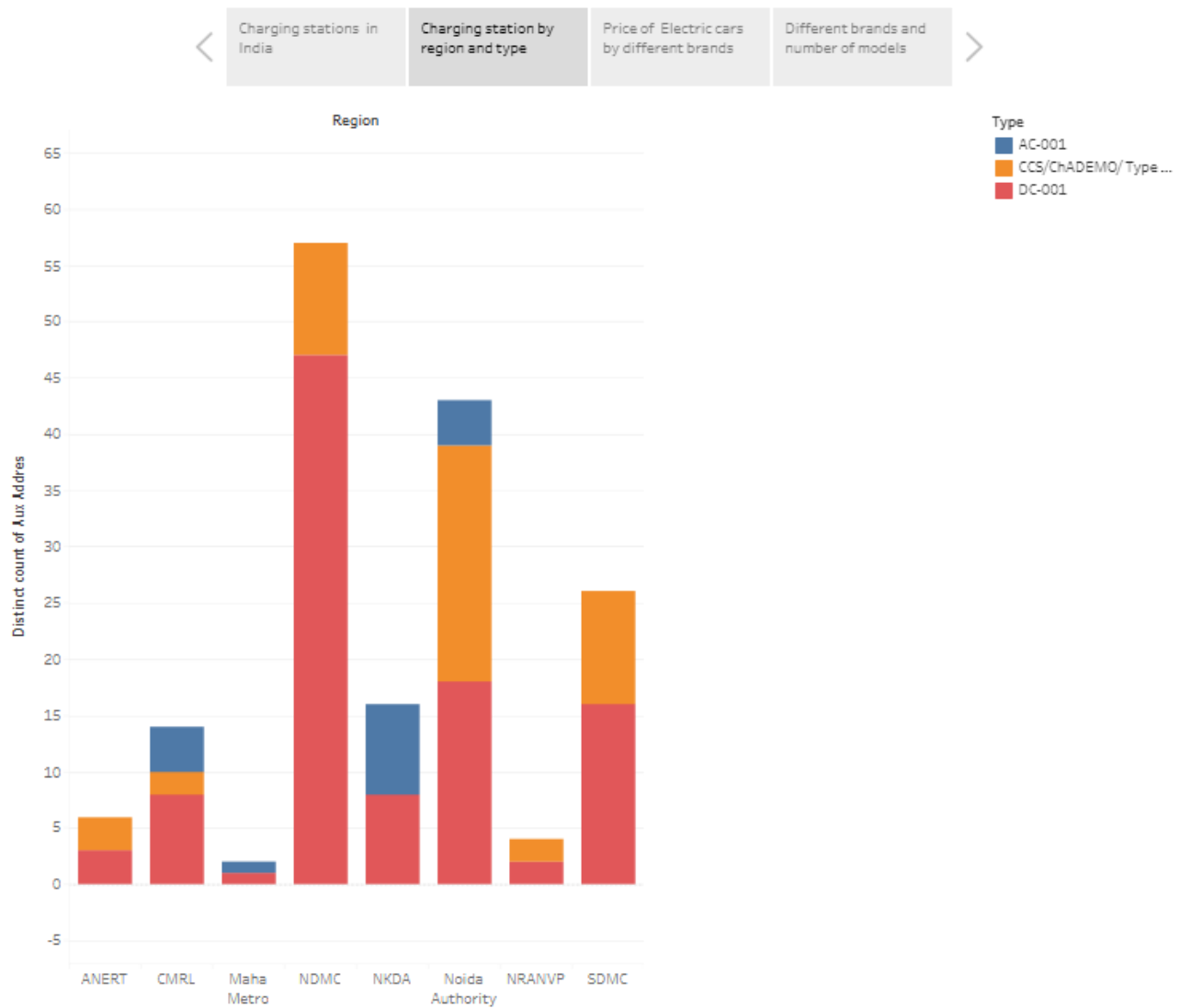


### Fig 11.1: Tableau Dashboard

## The Dashboard shows

- Charging Stations by region and type in India
- EV Charging stations map of India
- Different EV cars in India
- Top speed for different Brands
- Price for different cars in India
- Top 10 most efficient EV Brands
- Brands according to Bodystyle
- Brand filtered by PowerTrain type
- No of models by each brand
- Different brands of EV Cars globally
- Different brands of EV Cars in India

## Story of Electric cars in India



**Fig 11.2: Tableau Story**

The story shows

- Charging stations in India
- Charging Stations by region and type
- price of electric cars by different brands
- different brands and number of models

## Home Page

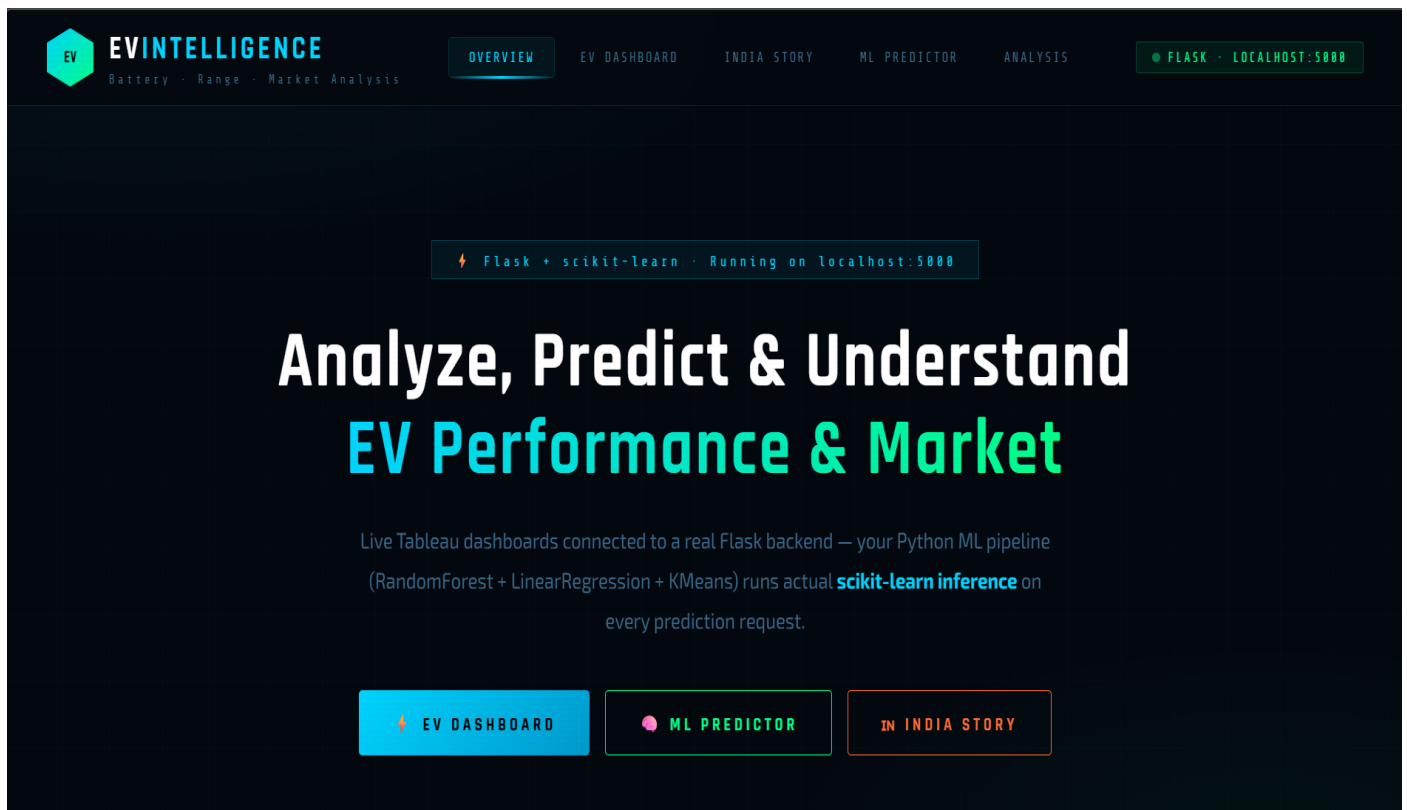


Fig 11.3: Home page of EV INTELLIGENCE

### Predict Electrical Vehicle range and efficiency

Predict Electrical vehicle range and efficiency using ML algorithms.

Using input features like speed ,battery capacity ,charging time and number of seats in the car.

Gives Quick Estimated Output to the user.

 **Configure EV Specs**

POST → /api/predict → Ensemble model inference

ACCELERATION 0-100 KM/H

6.5 S

1.5s

Sport

15s

TOP SPEED

165 KM/H

100

280

BATTERY CAPACITY

63 KWH

15

130

FAST CHARGE SPEED

300 KM/H

0 (none)

1050

NUMBER OF SEATS

5

2

5

9

DRIVETRAIN

AWD

▼

SEGMENT

C – Compact

▼

BODY STYLE

SUV

▼

RAPID CHARGE

Yes

▼

 **PREDICT NOW**

Fig 11.4: Inputs are given to EV by using ML models



Fig 11.5: Output is generated based on the inputs

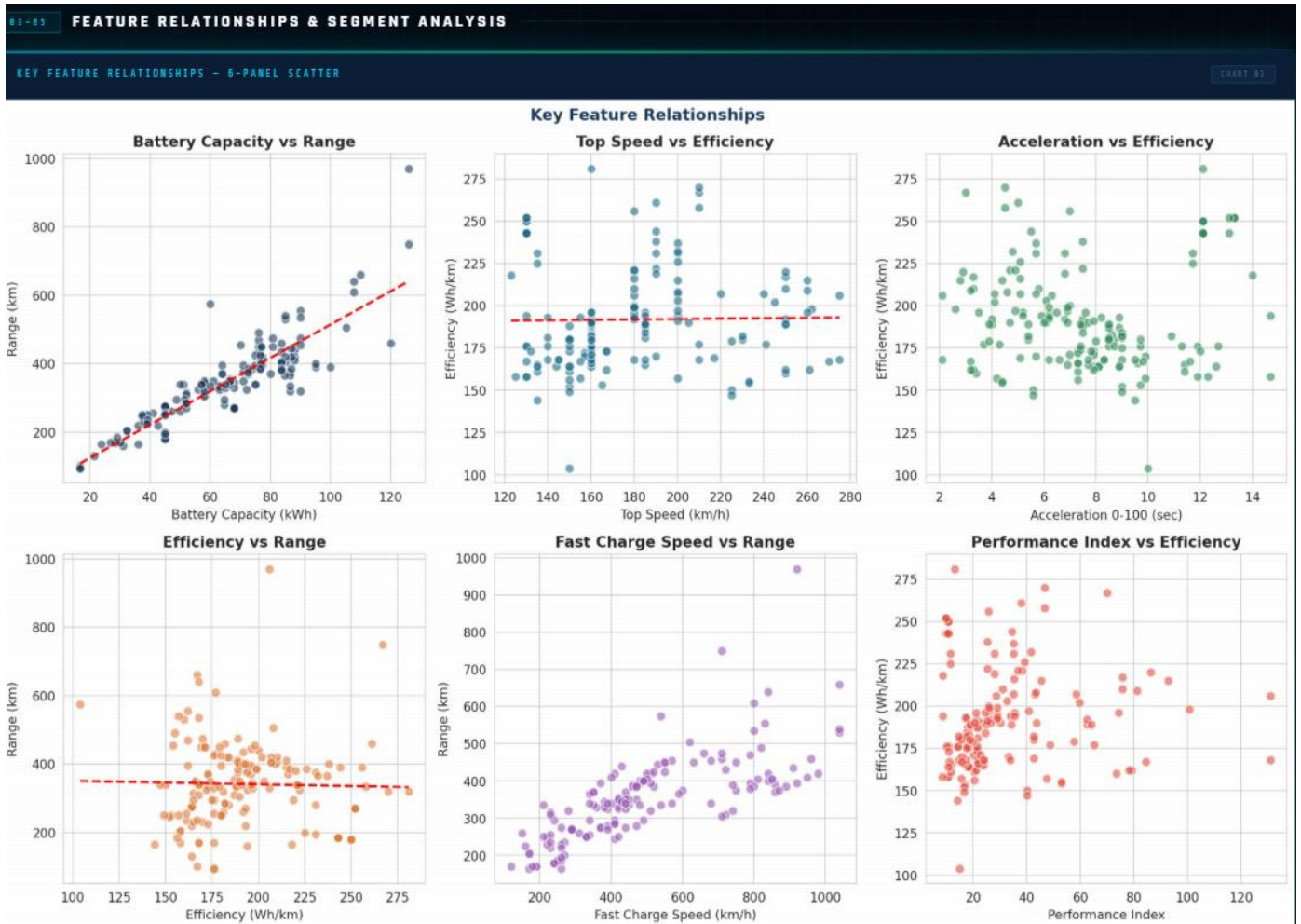


Fig 11.6: Relationship between key features of Electrical Vehicle

## **CONCLUSION**



# CONCLUSION

The “EV Battery Performance and Range Monitoring System” project was developed to analyze and understand the key factors affecting the performance, efficiency, and driving range of electric vehicles. By integrating multiple datasets related to electric vehicle specifications, battery capacity, pricing, and charging infrastructure, the project provided a comprehensive data-driven perspective on the current state of electric mobility.

Throughout the project, significant emphasis was placed on examining the relationship between battery capacity and driving range. The analysis demonstrated that while larger battery capacities generally contribute to extended range, overall vehicle efficiency is influenced by additional factors such as vehicle weight, motor power, and aerodynamic design. This highlights the importance of optimizing both battery technology and vehicle engineering to achieve improved energy utilization.

The study of charging infrastructure revealed that charging stations in India are unevenly distributed, with a higher concentration in metropolitan and economically developed regions. The limited availability of fast-charging stations in several areas continues to be a major challenge for long-distance travel and contributes to range anxiety among electric vehicle users. These findings emphasize the need for strategic infrastructure expansion to support widespread EV adoption.

Through the use of SQL databases, Tableau dashboards, and story visualizations, the project successfully transformed raw data into meaningful insights. Interactive dashboards enabled the comparison of vehicle brands, battery capacities, efficiency levels, and price variations, while the story feature provided a structured narrative that made the analysis more accessible and engaging. The integration of these visual analytics tools enhanced the interpretability of complex datasets and supported informed decision-making.

Performance testing confirmed that the system is capable of handling large volumes of data and rendering multiple visualizations efficiently without significant delay. The inclusion of web integration using Flask further demonstrated the practical deployment potential of the system by allowing users to access dashboards and stories through a browser-based interface.

Overall, the project achieved its primary objective of developing a comprehensive analytical platform to evaluate electric vehicle battery performance and driving range. The insights generated through this system can assist researchers, policymakers, and consumers in understanding the current limitations and future opportunities in the electric vehicle ecosystem. By combining data analytics, visualization, and web technologies, the project contributes to the broader goal of promoting sustainable and efficient transportation solutions.

## **FUTURE SCOPE**

# **FUTURE SCOPE**

The developed system provides a strong analytical foundation for evaluating electric vehicle battery performance, range efficiency, and charging infrastructure. However, with the rapid growth of electric mobility and advancements in data technologies, the system can be further enhanced with additional features and intelligent capabilities to improve accuracy, scalability, and real-world usability.

## **Advanced Data Analytics and Machine Learning Improvements**

Future versions of the system can incorporate advanced machine learning techniques to provide predictive insights into battery degradation, range estimation, and charging behavior. By integrating larger and continuously updated datasets from real-world vehicle usage, the system can generate more precise and dynamic predictions of vehicle performance under different driving conditions.

Possible enhancements include:

- Implementing deep learning models to predict battery health and range more accurately
- Using real-time vehicle telemetry data to improve prediction reliability
- Applying feature selection and optimization techniques to identify the most influential factors affecting range and efficiency
- Enhancing model evaluation using advanced performance metrics and cross-validation techniques

## **System and User Interface Enhancements**

The current web interface provides access to dashboards and story visualizations through a browser. In the future, the system can be expanded to support more interactive and user-centric features. A responsive and mobile-friendly interface would allow users to access vehicle analytics from smartphones and tablets, making the platform more accessible to a wider audience.

Future system improvements may include:

- Development of a mobile-optimized web interface for improved accessibility
- Implementation of user login and personalized dashboards for saving preferences and selected vehicles
- Addition of interactive comparison tools allowing users to compare multiple electric vehicles based on battery capacity, range, and efficiency
- Enhanced user interface design to provide smoother navigation and improved visual clarity

## **Integration with Real-Time and External Data Sources**

The system currently relies on structured datasets collected from public sources. In future, it can be integrated with real-time data feeds from electric vehicles, charging networks, and government

infrastructure databases. This would allow the platform to provide live updates on charging station availability, battery performance trends, and vehicle efficiency metrics.

Potential integration opportunities include:

- Connecting with public charging network APIs to display live charging station status
- Integrating with vehicle telematics systems to collect real-time battery and performance data
- Incorporating weather and traffic data to analyze their impact on vehicle range and energy consumption

### **Expansion of Visualization and Analytical Capabilities**

The dashboards developed in this project provide descriptive analytics and comparative visualizations. Future enhancements can include predictive and prescriptive analytics that help users make informed decisions about route planning, vehicle selection, and charging strategies.

Future visualization improvements may include:

- Predictive dashboards showing expected battery range based on driving conditions and terrain
- Map-based route optimization visualizations highlighting optimal charging stops for long-distance travel
- Time-series analysis of battery performance and degradation over extended usage periods
- Interactive simulations to estimate range variations under different load and climate conditions

### **Practical and Industrial Applications**

As electric vehicles become more common, such analytical systems can be adopted by various stakeholders in the transportation ecosystem. Automotive manufacturers, fleet operators, and policy planners can use the system to evaluate vehicle performance trends and infrastructure requirements.

Possible real-world applications include:

- Assisting electric vehicle manufacturers in improving battery management strategies
- Supporting fleet management companies in monitoring vehicle efficiency and optimizing operational costs

### **Scalability and Cloud Deployment**

The modular design of the system allows it to be easily deployed on cloud platforms in future implementations. Cloud integration would enable handling of larger datasets, improved processing power, and broader accessibility for users across different geographic locations.

Future cloud-related enhancements include:

- Deployment of the application on cloud services for improved scalability and availability
- Use of distributed data processing frameworks to analyze large volumes of EV telemetry data

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# BIBLIOGRAPHY

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# **APPENDIX 1**

## **KEY WORDS & ACRONYMS**

## APPENDIX 1: KEY WORDS & ACRONYMS

Keyword	Description
Tableau	Tool used for creating dashboards and story visualizations
Dashboard	Interactive charts showing EV performance and infrastructure data
Story	Sequence of visualizations explaining EV trends and insights
Data Visualization	Graphical representation of EV and charging station data
EV Dataset	Dataset containing EV specifications and charging infrastructure details
EV Performance Analysis	Evaluation of battery efficiency, range, and vehicle performance
Battery Capacity	Amount of energy stored in EV battery (kWh)
Driving Range	Distance an EV can travel on a full charge
Energy Efficiency	Measure of range per unit battery capacity (km/kWh)
Charging Infrastructure	Network of EV charging stations and their distribution
Data Preprocessing	Cleaning and preparing EV dataset for analysis
Feature Selection	Identifying key attributes affecting EV performance
Exploratory Data Analysis	Identifying patterns and trends in EV data
Price Distribution	Analysis of variation in EV prices
Correlation Analysis	Relationship between battery, range, and efficiency
Cost Efficiency	Price per kilometer of range analysis



Keyword	Description
Powertrain Type	Classification of EVs (BEV, Hybrid, etc.)
Brand Analysis	Comparison of EV manufacturers and their models
Flask	Web framework used for dashboard integration
Web Application	Interface to display dashboards and user interaction
User Interaction	Applying filters and exploring dashboards
Interactive Filters	Options to explore data based on user selection
Data Insights	Meaningful information derived from visual analysis
EV Market Trends	Patterns in EV adoption and infrastructure growth

## ACRONYMS

Acronym	Full Form
EV	Electric Vehicle
BEV	Battery Electric Vehicle
HEV	Hybrid Electric Vehicle
PHEV	Plug-in Hybrid Electric Vehicle
EDA	Exploratory Data Analysis
ML	Machine Learning
SQL	Structured Query Language

Acronym	Full Form
DB	Database
API	Application Programming Interface
UI	User Interface
kWh	Kilowatt-hour
km/kWh	Kilometers per Kilowatt-hour
BMS	Battery Management System
SOC	State of Charge
SOH	State of Health
EVSE	Electric Vehicle Supply Equipment
GPS	Global Positioning System
CSV	Comma Separated Values
HTML	HyperText Markup Language

## **APPENDIX 2**

### **RESEARCH PAPER**